

PROJECT REPORT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Automating ticket categorization, assignment and routing using a workflow-based solution

Project Domain: IT Service Management / Workflow Automation

Primary Platform: ServiceNow Flow Designer

Project Type: Automation and Intelligent Ticket Management

1. INTRODUCTION

1.1 Project Overview

Auto Ticket Classification Using Flow Designer is an IT service management automation project that classifies incoming support tickets according to their content, category, priority and assignment requirements. The solution uses Flow Designer to automate repetitive ticket-processing activities and route tickets to the appropriate support group.

When a ticket is created, the workflow evaluates the available ticket information such as short description, description, category, urgency and impact. Based on predefined classification rules, the flow updates the ticket with the appropriate category and assignment details and can trigger notifications or additional actions.

1.2 Purpose

The purpose of the project is to reduce manual ticket classification, improve routing accuracy, shorten response time and provide a consistent process for handling service requests and incidents. The automated workflow also helps service desk teams concentrate on resolving issues instead of performing repetitive administrative tasks.

2. IDEATION PHASE

2.1 Problem Statement

In a conventional service desk environment, newly created tickets may require manual review before they are categorized and assigned. This can cause delays, inconsistent categorization, incorrect routing and unnecessary workload for service desk personnel. A workflow-based automated classification mechanism is required to process tickets consistently and quickly.

2.2 Empathy Map Canvas

- **User says:** "My issue has been submitted, but I am not sure who will handle it."
- **User thinks:** "Will my ticket reach the correct team quickly?"
- **User does:** Creates a ticket and waits for acknowledgement or assignment.
- **User feels:** Uncertain when classification and assignment take too long.
- **Service desk team:** Wants fewer repetitive classification tasks and better routing consistency.

2.3 Brainstorming

- Use Flow Designer to trigger automation whenever a ticket is created or updated.
- Define classification rules using ticket fields and keywords.
- Automatically set category, subcategory, priority and assignment group where applicable.
- Send notifications after successful classification or assignment.
- Maintain logs and exception handling for tickets that cannot be confidently classified.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System Response
Ticket Creation	Submits incident/request	Ticket record is created
Initial Processing	Waits for processing	Flow is triggered automatically
Classification	Provides issue details	Rules evaluate ticket information
Assignment	Expects correct support team	Assignment group is selected
Notification	Receives update	Status/assignment notification is sent
Resolution	Works with support team	Ticket proceeds through normal lifecycle

3.2 Solution Requirement

- The system shall trigger a flow when a new ticket is created.
- The flow shall read relevant ticket fields such as description, category, urgency and impact.
- The workflow shall apply predefined classification rules.
- The workflow shall update the ticket category and assignment information.
- The system should support notification and exception handling.
- The solution should maintain a consistent and auditable workflow.

3.3 Data Flow Diagram

Ticket Created → Flow Trigger → Read Ticket Data → Apply Classification Rules → Determine Category/Priority → Select Assignment Group → Update Ticket → Notify/Log Result

3.4 Technology Stack

Component	Technology / Tool
Workflow Automation	ServiceNow Flow Designer
ITSM Platform	ServiceNow
Ticket Data	Incident / Request records
Business Rules	Flow conditions and decision logic
Notifications	ServiceNow notification actions
Testing	Flow execution, test records and logs

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution directly addresses the delay and inconsistency caused by manual ticket classification. By embedding classification logic in an automated Flow Designer workflow, tickets can be processed immediately after creation. This provides a repeatable process and reduces the number of manual steps required by service desk staff.

4.2 Proposed Solution

The proposed system consists of an automated flow that starts when a ticket satisfies the configured trigger condition. The flow retrieves the ticket information, checks classification conditions, determines the appropriate category and assignment group, updates the record and optionally sends a notification. Tickets that do not match the configured rules can be routed to a default queue for manual review.

4.3 Solution Architecture

Layer	Description
User Layer	End users create incidents or service requests.
ServiceNow Record Layer	Ticket information is stored in the relevant record.
Flow Designer Layer	Trigger, conditions, actions and decision logic execute.
Classification Layer	Rules determine category, priority and routing.
Assignment Layer	Ticket is assigned to the appropriate support group.
Notification Layer	Relevant stakeholders can be notified of updates.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Major Activity	Deliverable
1	Requirement study	Problem definition and requirements
2	Workflow design	Flow logic and classification rules
3	Configuration	Flow Designer implementation
4	Testing	Test cases and execution results
5	Validation	Classification and routing verification
6	Documentation	Final project report and demonstration

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focuses on whether the flow is triggered correctly, applies the expected classification rule, updates the ticket accurately and handles unmatched conditions safely. Test data should include common ticket categories, different urgency/impact combinations and tickets containing incomplete or unexpected information.

Test Case	Expected Result
New ticket matches Category A rule	Category A and configured assignment are applied.
New ticket matches Category B rule	Category B and configured assignment are applied.

Test Case	Expected Result
High urgency/impact ticket	Configured priority rule is applied.
No classification rule matches	Ticket is routed to default/manual-review path.
Ticket updated after creation	Flow executes only when the configured condition is met.

7. RESULTS

7.1 Output Screenshots

The final report should include screenshots showing the configured Flow Designer flow, trigger conditions, classification decision logic, ticket field updates, assignment group selection and sample ticket results. Screenshots should demonstrate the complete flow from ticket creation to automated classification and assignment.

Suggested screenshots to insert:

- Flow Designer main workflow canvas.
- Trigger configuration for new ticket creation.
- Classification conditions / decision logic.
- Ticket update action showing category and assignment fields.
- Sample ticket before automation.
- Sample ticket after automation with classification and assignment.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces repetitive manual classification work.
- Improves consistency in ticket categorization and routing.
- Provides faster processing of newly created tickets.
- Makes workflow logic easier to audit and modify.
- Supports scalable automation as additional rules are introduced.

Disadvantages / Limitations

- Rule-based classification depends on the quality of configured conditions.
- Unusual or ambiguous ticket descriptions may require manual review.
- Changes in service categories or support groups require workflow maintenance.
- Large rule sets can become difficult to maintain without clear documentation.

9. CONCLUSION

Auto Ticket Classification Using Flow Designer demonstrates how workflow automation can improve IT service management operations. By automatically processing ticket information, applying classification rules and routing tickets to the appropriate support group, the solution reduces manual effort and creates a consistent ticket-handling process. The project provides a foundation for extending

automation to additional service management activities.

10. FUTURE SCOPE

- Introduce AI/ML-based classification for more flexible understanding of ticket descriptions.
- Use historical ticket data to improve classification rules and identify recurring issues.
- Add confidence-based routing so uncertain tickets are sent for manual validation.
- Integrate automated knowledge-article recommendations.
- Add analytics dashboards for classification accuracy, routing time and exception rates.
- Expand automation to incident prioritization, SLA monitoring and escalation.

11. APPENDIX

Source Code (if any)

Include Flow Designer configuration details, scripts or supporting implementation files, if used.

Dataset Link

Insert the project dataset or sample ticket-data location used for testing. If no external dataset is used, state that sample ServiceNow ticket records were used.

GitHub & Project Demo Link

Insert the GitHub repository and project demonstration link when available.